

Vipransh Ojha

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Education

VIT Bhopal University

B.Tech in Computer Science and Engineering (AI & ML Specialization)

Bhopal, India

Sep 2023 – May 2027

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, C++, Java, SQL, HTML5, CSS3
- **AI/ML Frameworks:** TensorFlow, PyTorch, OpenCV, Scikit-learn, Keras, Computer Vision, NLP, Deep Learning
- **Web Development:** React, Flask, FastAPI, Node.js, Express.js, Tailwind CSS, Bootstrap, REST APIs
- **Database & Tools:** MongoDB, PostgreSQL, SQLite, Git, Docker, Power BI, AWS Cloud Services, Linux, MediaPipe
- **Specialized Libraries:** OpenAI APIs, Google TTS, Whisper, NEAT, Pygame, Pandas, NumPy, Matplotlib

Experience

Software Development Intern

BISAG-N (Government of India)

New Delhi, India

May 2025 – Jun 2025

- Built Discover Places smart location recommendation system serving 500+ users with 85% recommendation accuracy using machine learning algorithms.
- Implemented comprehensive Flask backend architecture with PostgreSQL database for real-time data processing and geospatial analysis.
- Integrated OpenStreetMap APIs for dynamic mapping functionality and developed admin management system with user authentication.

Projects

Forensic Identification using Dental Radiology

AI/ML Research Project

Jul 2025 – Present

Python, Computer Vision, Deep Learning, TensorFlow

- Developed AI-powered forensic identification system achieving 92% accuracy on 1,000+ dental radiography samples using convolutional neural networks.
- Implemented advanced computer vision and deep learning models for automated dental X-ray analysis with comprehensive image preprocessing.
- Collaborated with forensic experts and law enforcement professionals ensuring practical applicability and regulatory compliance standards.

SpeechSync - Real-Time Multilingual Translation System

Collaborative AI Project

Aug 2024 – Dec 2024

Python, OpenAI Whisper, Neural Machine Translation

- Co-developed comprehensive speech-to-speech translation system supporting 10+ languages with 85% accuracy and sub-2 second latency.
- Implemented modular pipeline architecture using OpenAI Whisper for speech recognition and Google Text-to-Speech APIs for voice synthesis.
- Built scalable real-time processing framework handling concurrent users with optimized memory management and error handling.

Computer Vision Projects Suite

Human-Computer Interaction — AI/ML

Jan 2024 – Jun 2024

Python, OpenCV, MediaPipe, Computer Vision

- Developed Air Writing Recognizer with 90% accuracy for real-time gesture-based text input using advanced computer vision algorithms.
- Built Gaze-Controlled and Gesture-Controlled Mouse systems with sub-50ms response time enabling hands-free navigation solutions.
- Implemented multi-modal interaction system combining gesture recognition and voice controls for comprehensive accessibility applications.

Co-Curricular and Extra-Curricular Activities

- Active contributor to open-source AI projects with focus on social impact, accessibility solutions, and community-driven technology initiatives.
- Participated in coding competitions and hackathons demonstrating advanced problem-solving skills in competitive programming environments.
- Engaged in AI research activities exploring cutting-edge technologies and contributing to innovative solutions in machine learning domain.

Additional Information

- **Languages:** Fluent in English and Hindi with professional proficiency in technical communication and documentation.
- **Interests:** Cricket, Badminton, Swimming, Music composition, AI Research, complex algorithmic problem solving, and emerging technology trends.